

BOAMP Digital Procurement Study

Successor Linkage, Survival, and Trend Methodology

BOAMP Data Science Internship Project

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Technical Summary

This report states the BOAMP successor-linkage, survival, and trend methodology and the results it produces. The study does not claim to certify legal contract renewals. Its event is an *observable successor procurement*: a later BOAMP procurement episode from the same buyer that is sufficiently similar to an earlier awarded digital procurement episode.

The Grand Ouest survival cohort contains 3,800 awarded digital procurement episodes. The main linkage rule accepts 544 successor links, giving an observed event rate of 0.143 and a median observed successor time of 31.82 months among linked events. The candidate pool is broad: 763,417 candidate pairs from 3,520 anchors with at least one candidate.

Linkage accuracy is read against the Grand Ouest regional reference: a stratified review of 120 anchors of which 88 are usable, carried out against real BOAMP notices before these methods existed. On its locked split of 72 anchors (18 with a reviewed successor), `M_B_text_ranking @ 0.70` has precision 0.875 (95% CI 0.529–0.978), recall 0.389 (95% CI 0.203–0.614), false-positive rate 0.000, and 8 accepted links. Recall is capped at 0.913 by candidate generation, which reaches 21 of the 23 reviewed successors. These are reference-sample estimates on fewer than a hundred anchors, not independently validated accuracy: the labels were generated by a single LLM research pass and spot-checked on a subset by the project owner, not verified anchor-by-anchor and not reviewed by an independent specialist panel. `M_C_weighted_gated` recovers more positives (recall 0.667) but admits more false positives (FPR 0.185). This trade-off is important because a false positive in a survival dataset creates both a false event and a false event time, while an abstained link is handled conservatively as right-censoring.

1 Context

Gigalis is the digital central purchasing body (*centrale d’achat numérique*) for public institutions in the Pays de la Loire region and beyond. It designs pooled framework agreements in cloud, cybersecurity, digital infrastructure, and artificial intelligence, and its strategic relevance depends on anticipating members’ future needs. The internship’s central question is: using historical public digital procurement data, how can the probability that a contract or technological segment generates an identifiable purchasing need within the next 12–24 months be estimated, and how can that estimate inform Gigalis’s central purchasing strategy?

This decomposes into three sub-problems: a *lifetime problem* (survival analysis: how long between a digital procurement episode and its successor?), a *trend problem* (change-point detection: which technological segments are growing, plateauing, or declining?), and a *text-signal problem* (NLP: extracting a contract’s technological theme automatically). This project answers the first two in depth; the NLP sub-problem is addressed with a reproducible coarse substitute rather than a trained classifier (§8).

BOAMP does not encode legal contract renewal as an explicit field, so the study cannot certify legal renewals. It instead constructs and evaluates a proxy: an *observable successor procurement*, a later BOAMP episode from the same buyer that is sufficiently similar to, and temporally plausible relative to, an earlier awarded digital procurement episode. Every downstream result – survival curves, hazard ratios, trend segments – is conditioned on this proxy, not on certified legal ground truth. The strongest defensible framing is: because legal renewal status is not directly observed, the study measures observable successor procurements under a precision-first linkage rule, then assesses how sensitive its descriptive conclusions are to that rule.

2 What The Pipeline Measures

Let i index an earlier awarded procurement episode and j index a later candidate episode. The object of interest is not a legal renewal certificate, because BOAMP notices do not provide

that ground truth consistently. Instead, the pipeline estimates whether the data show a later procurement episode that is credible enough to be treated as a successor:

$$Y_i = \begin{cases} 1, & \text{if one accepted observable successor is found before the cutoff,} \\ 0, & \text{otherwise.} \end{cases}$$

The event time is

$$\tau_i = C_{\hat{j}} - A_i,$$

where A_i is the award-date origin of episode i , $C_{\hat{j}}$ is the first-publication date of the accepted successor, and $\hat{j}$ is the selected candidate. If no accepted successor is found before 2025-12-31, the episode is right-censored.

3 End-to-End Pipeline

The implemented pipeline is:

BOAMP notices $\rightarrow$ standardised notices $\rightarrow$ procurement episodes
 $\rightarrow$ candidate pairs $\rightarrow$ linkage scoring $\rightarrow$ one successor or abstention $\rightarrow$ survival dataset.

Standardisation and feature engineering. The preprocessing stage standardises dates, buyer identifiers, buyer names, CPV codes, text fields, procedure information, framework flags, and duration fields where they are explicitly available. It does not impute missing duration or invent expected expiry dates. Buyer standardisation is deliberately conservative: municipal variants such as commune/ville/mairie can be harmonised, but intercommunal legal forms are preserved as distinct entities, and conflicting validated SIRENs block a buyer match. The legal-form audit reports 0 accepted links with conflicting validated SIREN evidence and 0 accepted municipal/intercommunal mixes.

Episode reconstruction. BOAMP is notice-level data, but the analysis needs procurement-level events. Therefore notices belonging to the same procurement are reconstructed into an episode. This avoids treating a consultation notice, correction notice, award notice, or lot-level administrative notice as separate renewals.

Candidate generation. Candidate generation is intentionally broad and is not the final model. For an anchor episode with award date A_i and candidate publication date C_j , the candidate is exposed only if

$$A_i + 90 \leq C_j \leq A_i + 2920.$$

The lower bound removes very near follow-up notices and parallel administrative activity. The upper bound is approximately eight years; it keeps the candidate pool inclusive enough for long public-procurement cycles. Precision is then controlled by the selection stage rather than by a narrow time window. The blocking rule generates 763,417 candidate pairs, with a median of 89 candidates per anchor.

Candidate generation is deliberately recall-oriented, and it is a separate stage from final linkage. A candidate episode must satisfy buyer-identity plausibility and the broad future-time window above, but exact CPV continuity is *not* imposed as a hard blocking rule, because CPV coding is incomplete, often generic, and assigned by the contracting authority rather than validated. Retaining plausible successors is the blocking stage’s job; final precision is controlled by the linkage scorer and the frozen acceptance threshold. In record-linkage terms this quantity is *pairs completeness*, the share of true matches surviving the blocking step, which [Christen and](#)

Goiser (2007) identify as a confounder that must be published alongside any linkage-quality figure rather than folded into it.

Relaxing hard same-CPV blocking is checked against the reference rather than asserted. Among the 23 reviewed successor pairs – labelled with no knowledge of any linkage method – 9 (39.1%) connect episodes in *different* CPV divisions. A hard same-division block would therefore discard those 9 reviewed successors outright and cut the attainable recall ceiling to 0.609. The discarded cases are substantive continuations rather than noise: equipment purchases followed by their maintenance contracts (CPV-32 or CPV-35 to CPV-50) and software purchases followed by the corresponding service contracts (CPV-48 to CPV-72). This is the ordinary goods/services split built into the CPV hierarchy itself, compounded by assignment error that Siciliani et al. (2023) report is frequent even for human experts given the size of the vocabulary.

4 Grand Ouest Regional Reference

Linkage evaluation uses a Grand Ouest regional reference sample drawn from the study population itself: 120 awarded digital procurement anchors stratified by CPV theme, buyer-identifier quality, and duration availability across Bretagne, Pays de la Loire, and Normandie. Each anchor was reviewed on 2026-08-11 against the notices and official BOAMP URLs of its candidates. The review is a single LLM research pass over those notices, their official URLs, and wider public sources, spot-checked on a subset by the project owner; it proposes one successor or an abstention per anchor. No linkage method existed or was consulted when the labels were produced, so the labels are independent of every method scored against them.

Reference anchors are resolved onto the procurement-episode table through their BOAMP notice identifiers. 8 anchors did not resolve to exactly one episode and were dropped rather than guessed, and anchors the review declined to decide are excluded rather than counted as negatives, leaving 88 evaluable anchors.

The sample is split into a pilot part and a locked part. The acceptance threshold was fixed before the locked part was read, which is what allows locked-split figures to be reported as held out. The pilot split carries 16 evaluable anchors (5 with a reviewed successor) and 5,221 pair rows; the locked split carries 72 evaluable anchors (18 with a reviewed successor) and 20,917 pair rows.

Two anchor counts appear for the locked split and the difference is by construction, not a discrepancy: 72 anchors are evaluable at anchor level, of which 69 also have at least one exposed candidate pair and so appear in the pair-level table. The remaining 3 generated no candidate at all, which is a blocking-stage loss counted against recall rather than a scoring failure. Anchor-level metrics use 72; pair-level ROC and precision-recall curves use 69.

Candidate generation, not scoring, sets the ceiling on recall against this reference. The exposed candidate pool contains 21 of the 23 reviewed successors, so no method evaluated here can exceed a recall of 0.913; the gap is a blocking-stage limitation rather than a scoring failure. The wording matters: candidate generation *exposed* 21 of 23 reviewed successors in this sample. That is a property of the reference, not an estimate of population recall.

Both unreachable cases were attributed to the specific blocking condition that rejected them, evaluated in the order the generator applies them, and neither is an implementation defect. In the first, the reviewed anchor never reached the blocking step at all: its episode carries an award notice, but that notice has no structured Grand Ouest address, so it is absent from the regional notice table, no award date could be resolved, and the episode was dropped from the cohort – one of the 26 episodes the cohort funnel already counts under that heading. In the second, the buyer changed legal form between the two procurements, from a communal social action centre (CCAS) to its intercommunal successor (CIAS); the anchor carries no SIREN, so no identifier could bridge the transition, and the normalised names differ. Buyer blocking rejected the pair by

design, because the pipeline deliberately preserves intercommunal legal forms as distinct entities rather than merging them on name similarity.

That second case is the known weak point of buyer identification in French procurement notices rather than a defect specific to this pipeline: [Potin et al. \(2023\)](#) report missing agent identification as the most serious quality problem in the French data, with buyer SIRETs populated on a minority of lots, and note that the same authority appears under alternative and former names. Recovering this particular link would require an external legal-succession register, not a change to the blocking rule. Both losses are accepted and documented rather than repaired; neither is repaired by loosening the rule, and the audit reports 0 unexplained cases, which is the check that distinguishes a blocking trade-off from a bug. The attribution is regenerated by `scripts/audit_candidate_generation.py`.

Five limitations bind everything computed from the regional reference. The labels were spot-checked on a subset rather than verified anchor-by-anchor; anchors outside that subset carry the model’s judgement as recorded, so this is a reference sample and not ground truth, and it is not an independent human specialist panel. The sources behind each individual label were not recorded, so a given anchor’s evidence trail cannot be fully reconstructed or independently re-executed. Negatives are corpus-relative: roughly 25 candidates per anchor were considered rather than the full pool, so a false-positive rate computed on them is conservative by construction and is a diagnostic on this sample rather than a population-wide rate. Judging whether later notice text continues an earlier need draws on the same text, CPV, and date evidence the text-ranking method scores, so the labels are method-independent without being fully evidence-independent. And the sample is small enough that every point estimate needs its interval read beside it.

5 Linkage Algorithms

All methods operate on the same exposed candidate set. This is crucial: the primary method is not comparing text over the whole BOAMP universe. Buyer and time plausibility are imposed before text ranking.

M_A : deterministic evidence. Define B_{ij} as buyer-identity support, P_{ij} as CPV continuity, and T_{ij} as text similarity. The deterministic rule accepts a link only when strong buyer evidence is present, CPV continuity is positive, and a minimum text signal is present:

$$B_{ij} = 1, \quad P_{ij} > 0, \quad T_{ij} \geq t_A.$$

It is interpretable, but it loses recall when CPV or buyer identifiers are missing or noisy.

M_B : text ranking. For each episode, the text fields are converted to TF-IDF vectors x_i and x_j . Text similarity is cosine similarity:

$$T_{ij} = \cos(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\| \|x_j\|}.$$

Within the same-buyer, plausible-time candidate set, the method selects

$$\hat{j}_i = \arg \max_j T_{ij},$$

and accepts the selected candidate only if

$$T_{i\hat{j}_i} \geq 0.70.$$

This is the primary method because it is simple, reproducible, auditable, and best matches the precision-first objective.

M_C : weighted gated score. This method combines evidence components into a score:

$$S_{ij} = 0.50B_{ij} + 0.25T_{ij} + 0.20P_{ij} + 0.05G_{ij},$$

where G_{ij} is a timing-plausibility score. Missing components are handled by the implemented score-normalisation logic rather than imputation. The method then applies gates and ranks by S_{ij} . It is useful as a higher-recall contrast, but the validation results show a materially higher false-positive rate.

M_D : Fellegi–Sunter probabilistic linkage. Let γ_{ij} be the discretised comparison vector for buyer, text, CPV, and timing evidence. Fellegi–Sunter estimates the likelihood of observing γ_{ij} under a match class M and a non-match class U , then uses the log-likelihood ratio

$$w_{ij} = \log \frac{P(\gamma_{ij} | M)}{P(\gamma_{ij} | U)}.$$

The fitted model provides `fs_match_weight` and `fs_match_probability`. On this cohort it does not outperform the simple text-ranking rule, likely because true successors are rare and same-buyer procurement activity contains many non-renewal lookalikes.

Declared duration. Declared contract duration is not imposed as a mandatory successor-linkage condition. Reliable duration is unavailable for most episodes, so a duration-conditioned rule could differentiate itself only on a minority of the cohort; and where the evidence does exist, the observed relationship between declared duration and successor timing is too weak to support a hard expected-expiry rule – many declared successors are published well before the declared end date. Duration is therefore retained as a descriptive diagnostic about data reliability, not as an event-definition criterion, and the descriptive comparison between declared duration and observed successor delay remains part of the evidence on that basis.

Sensitivity framework. Event-definition sensitivity is carried by four threshold and method arms ($M_B@0.80$, $M_B@0.70$, $M_B@0.60$, $M_C@0.70$), which vary the decision rule along the dimension that actually moves the event set, together with the borderline-band check described in §6.

6 Survival Modeling

The accepted-link decision from M_B defines the survival event. For episode i , τ_i is the time from award date A_i to the accepted successor’s publication date if $Y_i = 1$, or to the study cutoff (2025-12-31) if $Y_i = 0$ (administrative right-censoring). $Y_i = 0$ means no accepted observable successor was found before the cutoff; it is not proof of abandonment.

Kaplan–Meier. The non-parametric survivor function $S(t) = P(T > t)$ is estimated by [Kaplan and Meier \(1958\)](#)’s product-limit estimator. It is reported overall and stratified by CPV segment, which is the only stratification this study estimates; group differences use a multivariate log-rank test across those segments. Region and framework status enter the Cox model as covariates but are not estimated as separate Kaplan–Meier strata.

Cox proportional hazards. The semi-parametric model is

$$h(t | X) = h_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p),$$

with covariates selected for substantive relevance and data quality rather than automated search: CPV digital segment, buyer region, framework-agreement flag, validated-SIREN availability, and

centered award year. The rule of one covariate per ten observed events (Van Belle et al., 2002) is respected with 544 events supporting 8 covariates. The proportional-hazards assumption is tested with Schoenfeld residuals (Grambsch and Therneau, 1994); violations are reported and interpreted descriptively rather than silently dropped or used to discard the model.

Parametric models. Exponential, Weibull, log-logistic, log-normal, and generalized-gamma models are compared by log-likelihood, AIC, and BIC and checked graphically against the Kaplan–Meier curve. Their role is to identify the best-fitting family and to provide the instrument any extrapolation past the observation window would require. They are *not* the source of the reported 12/24-month probabilities: every horizon quoted in this report falls inside the observed window, and the smooth families flatten the empirical renewal shoulder, so the operational conditional probabilities are read off the Kaplan–Meier estimator, which imposes no shape.

Temporal validation. The model is fit once on episodes awarded 2015–2021 and scored out of time without refitting. The primary evaluation window is 2022–2024, as specified by the internship guideline; 2022–2025 is carried as a sensitivity read that adds the shortest-follow-up award cohort. Harrell’s concordance index C is reported on each split to assess discrimination and out-of-time stability, not to target a specific value.

Evaluation strategy. The three checks applied to the linkage – scoring it against a reference subset, comparing linked with unlinked episodes, and re-running the analysis under alternative linkage rules – follow the guidance of [Harron et al. \(2017\)](#) for evaluating linkage quality in linked-data analyses. That source supports the strategy; it does not validate the precision, recall, or threshold reported here.

Borderline-link robustness. Threshold sensitivity swaps one event definition for another; a separate check asks how much rests on the anchors the frozen rule nearly classified the other way. Anchors whose best candidate scores within ± 0.05 of the 0.70 acceptance bar are dropped entirely – borderline acceptances and borderline abstentions alike – and the headline Kaplan–Meier levels and hazard ratios are recomputed on the remainder. The band is fixed a priori and is not searched over.

Linkage sensitivity. Because the event is linkage-conditioned, the same Kaplan–Meier and Cox analyses are repeated under the strict ($M_B@0.80$), main ($M_B@0.70$), looser ($M_B@0.60$), and high-recall contrast ($M_C@0.70$) event definitions. A conclusion is reported as robust only when it is stable in sign and approximate magnitude across these arms.

7 Trend And Change-Point Detection

Quarterly awarded-episode counts $N_{s,q}$ are built for the overall cohort and each CPV digital segment from 2015Q2 (the first complete quarter) through 2025Q4, including zero-count quarters.

Change-point detection (PELT). [Killick, Fearnhead and Eckley \(2012\)](#)’s PELT algorithm minimizes a penalized segmentation objective on the z-standardized series with penalty $\beta = \lambda \log(n)$; the central result uses $\lambda = 1$ with sensitivity at 0.5 and 2.0, and a break is called stable only if it lies within one quarter under all three penalties.

Stationarity. Augmented Dickey–Fuller (unit-root null) and KPSS (level-stationary null) tests are run on each segment’s series. The two tests have opposite null hypotheses and are read together, not individually.

Regime detection (HMM). A 3-state Gaussian hidden Markov model is fit on the quarter-over-quarter *change* in episode count – not the level – for the overall cohort and the two highest-volume CPV segments, so states describe typical period-over-period direction (decline, plateau, growth) rather than absolute activity level. The Viterbi-decoded state sequence gives the current-quarter regime and its posterior probability. This complements PELT (which finds discrete historical breaks) with a continuously updated regime read; the two are not required to agree, and disagreement is reported honestly rather than reconciled.

Recent direction. An OLS slope over the latest 12 quarters gives an `increasing/decreasing/stable_or_uncertain` label at exploratory $\alpha = 0.10$, uncorrected for multiple testing. None of these methods forecasts future values or identifies the cause of a break; causal attribution requires documentary or stakeholder evidence not available in BOAMP alone. Monetary and duration trend series are omitted because no canonical awarded-amount field is validated at episode grain, and 2025’s duration-field completeness jump is a measurement change, not a genuine shift in contract durations.

8 Technology Segmentation

The internship guide’s L2 deliverable specifies a supervised technology taxonomy: 300–500 manually annotated contracts across 8–12 classes, a TF-IDF+logistic-regression/SVM baseline evaluated by macro-F1 and confusion matrix, and an optional CamemBERT comparison. **This is outside the scope of the reproducible analytical branch reported here.** A defensible L2 corpus requires two independent annotators and a Cohen’s kappa agreement statistic, and a second qualified annotator was not available; single-pass AI-assisted labels whose agreement was called "kappa" would carry exactly the self-consistency problem this project documents for its own reference labels (§4) rather than avoid it.

A finer technology-classification effort exists in parallel, outside this pipeline: `data/reference/technology_classification/` holds a 945-row export dated 2026-08-12 carrying a `Domaine` label per notice. It is read by no stage of this pipeline. It covers present-day national opportunities rather than the 2015–2025 Grand Ouest study cohort, and it arrives without the training corpus, the annotation guidelines, or the validation artifacts that would let its labels be audited or propagated over that historical cohort. This is a statement about what can be validated and applied historically here, not a judgement on the parallel work.

Every technology segment referenced in this report – cohort selection, Cox covariates, quarterly trend series – therefore uses CPV divisions 32 (telecommunications equipment), 35 (security), 48 (software), and 72 (IT services) as a reproducible substitute. This is a real scope reduction, not a disguised classifier: CPV divisions are official, zero-missingness EU categories under Regulation 213/2008, but they are coarser than a learned taxonomy and cannot distinguish sub-themes such as cloud versus on-premise infrastructure within a division. Completing L2 would require a second qualified annotator and a blinded double-annotation design of the kind recorded in `INDEPENDENT_LINK_REVIEW_PROTOCOL.md`.

9 Pilot Reference Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	3	0.667	0.400	0.091	0.188
M_B_text_ranking	70.000	2	0.500	0.200	0.000	0.125
M_C_weighted_gated	70.000	5	0.800	0.800	0.000	0.312
M_D_fellegi_sunter	65.000	2	0.000	0.000	0.091	0.125

Table 1: Unweighted metrics on the pilot split of the regional reference.

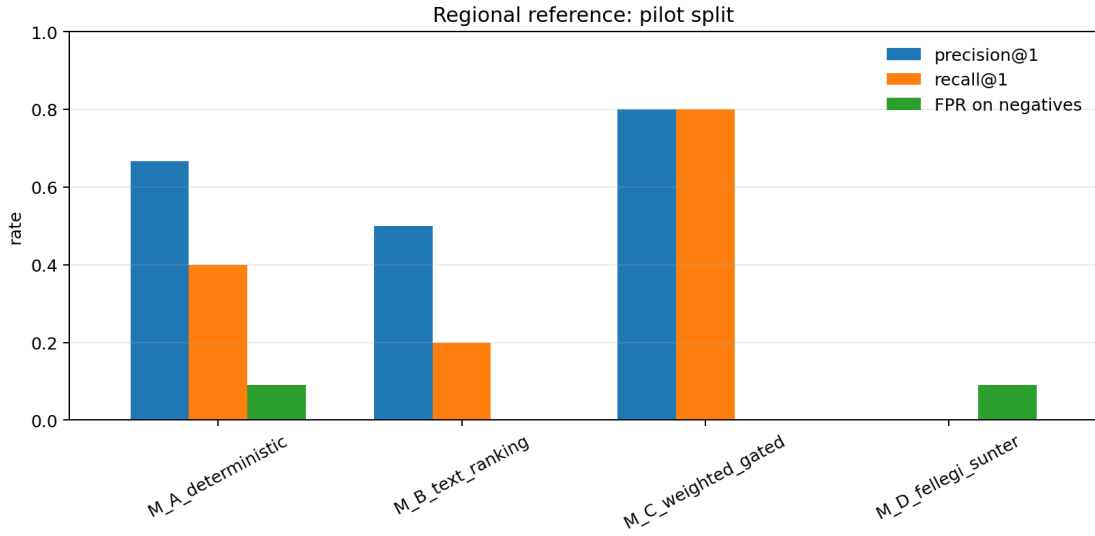


Figure 1: Pilot split method comparison generated from the regional-reference evaluation JSON.

10 Locked Reference Results

The table below is an internal method-comparison diagnostic. It must not be presented as external validation. The primary comparison is anchor-level exact-successor performance: each anchor can produce one accepted successor or abstain.

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

For this project, false-positive control is prioritised because false links fabricate survival events.

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	15	0.533	0.444	0.130	0.208
M_B_text_ranking	70.000	8	0.875	0.389	0.000	0.111
M_C_weighted_gated	70.000	23	0.522	0.667	0.185	0.319
M_D_fellegi_sunter	65.000	5	0.200	0.056	0.018	0.069

Table 2: Unweighted held-out metrics on the locked split of the regional reference.

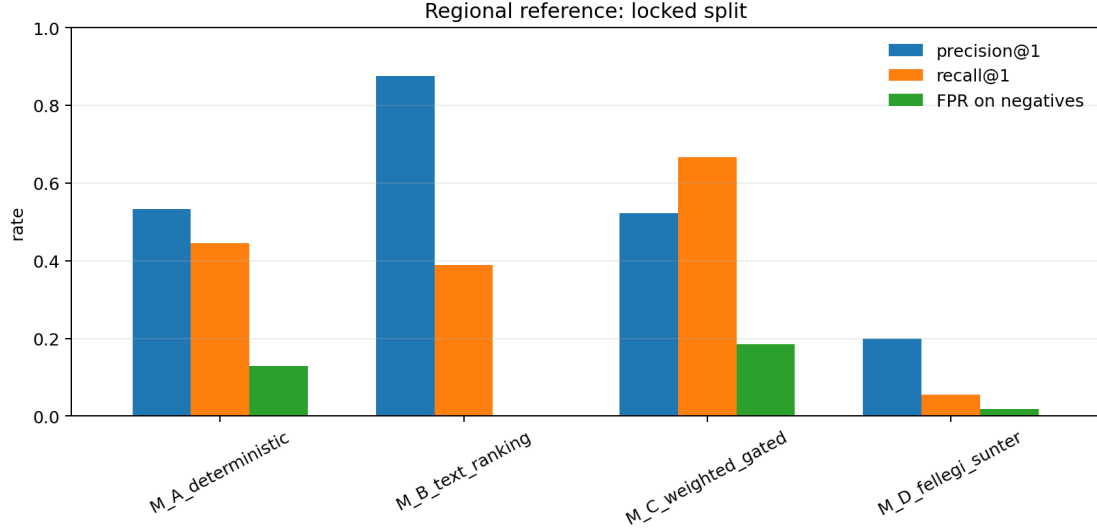


Figure 2: Held-out method comparison generated from the regional-reference evaluation JSON.

11 Quality Evidence And Interpretation

The held-out internal result supports retaining a conservative baseline for continued work; it does not establish final accuracy. M_B has only 8 accepted held-out links, so its precision estimate is necessarily sample-sensitive; one changed decision would move the number materially. Within this reference, the direction is coherent: M_B gives the best precision and the lowest false-positive rate among useful methods, while M_C shows the expected precision-recall trade-off.

The pair-level ROC and precision-recall curves should be read as score-ranking diagnostics, not as the final operating decision. The final pipeline decision is anchor-level: choose one successor or abstain. The ROC curve is stepped rather than smooth because the held-out reference contains a finite number of labelled pairs and many tied or near-tied scores.

The emphasis on precision-recall for rare positive decisions is supported by [Davis and Goadrich \(2006\)](#) and [Saito and Rehmsmeier \(2015\)](#). These sources justify the diagnostic choice; they do not validate the reference labels, numerical results, or selected threshold. Generic web illustrations are therefore treated as presentation aids only, not as academic evidence.

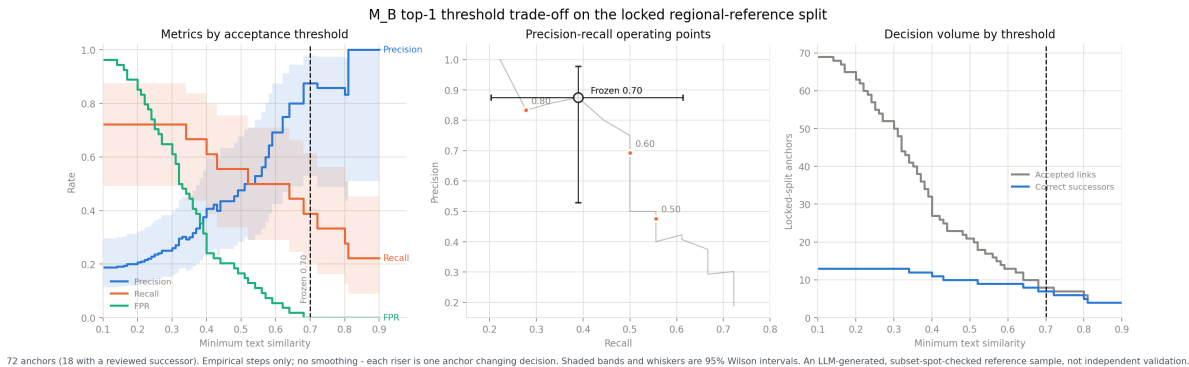


Figure 3: Project-specific, unsmoothed anchor-level threshold trade-off for M_B. The figure is empirical evidence from the locked split of the regional reference, not an idealised illustration.

On the locked split, 0.70 accepts 8 links of which 7 are correct (precision 0.875, recall 0.389, FPR 0.000), while 0.60 accepts 13 of which 9 are correct (precision 0.692, recall 0.500, FPR

0.056). On the pilot split the same two points give precision 0.500 and 0.667 respectively. The threshold was fixed before this reference was read, which is the only reason the locked split can be reported as held out; selecting one from these rows now would convert it into a tuning set. The lower threshold is therefore reported as a sensitivity arm rather than promoted after inspecting validation.

12 Model-Assisted Challenge Review

A separate blinded challenge review sampled 20 accepted links, 20 high-similarity structural-negative candidates, and 20 buyer-declared relationships. Its provenance is model-assisted rather than independent human specialist review. Among the 20 accepted links, 14 were confirmed, 5 were rejected, and 1 was uncertain. Conservative reviewed precision was 0.700, with exact 95% interval [0.457, 0.881]. This diagnostic falls below the 0.80 point target and exposes useful failure modes. It does not establish independent human validation, but it is sufficient to justify keeping the reported claim narrow and the lower threshold unpromoted.

13 Modeling Tables

The modeling-ready tables include strict, primary, broad, and non-match target columns, plus observable features. They include the Fellegi–Sunter columns `fs_match_weight` and `fs_match_probability`, so modeling and evaluation use the same feature state.

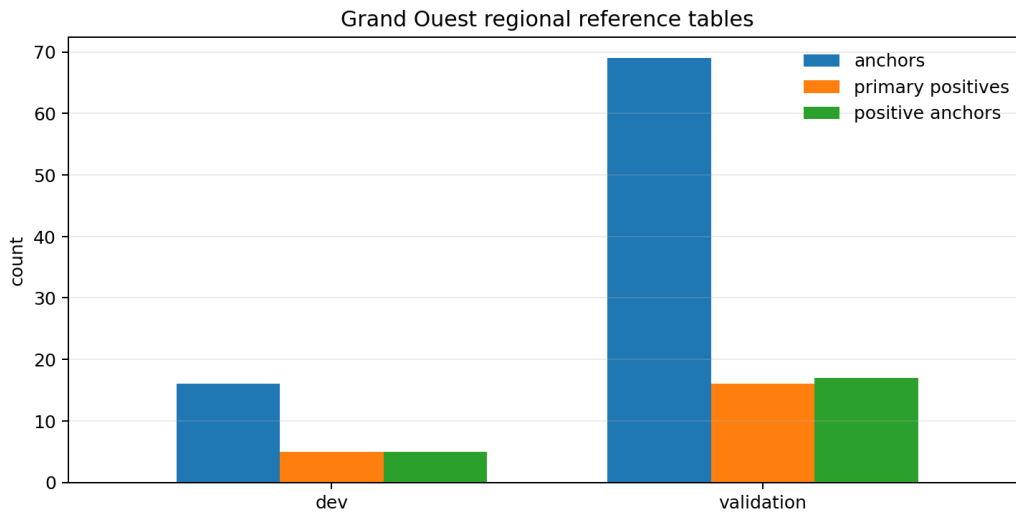


Figure 4: Regional-reference modeling table sizes and label support.

14 Survival Results

The survival dataset uses the main accepted links as events and all other episodes as right-censored observations at the study cutoff. The main variant contains 544 events and 3,256 censored observations (14.316% event rate). Segment event rates are 0.132 for CPV-32, 0.204 for CPV-35, 0.117 for CPV-48, and 0.143 for CPV-72. CPV-35 has the highest observed successor rate in the main arm.

Kaplan–Meier. The estimated probability of an observable successor is 4.621% by 12 months, 6.733% by 24 months, and 17.535% by 60 months. The Kaplan–Meier median survival time is

not reached: the curve never falls below 0.5 within the observation window, so no median is reported. This is distinct from the 31.82-month median delay *among linked events only*, which conditions on the event having occurred. A multivariate log-rank test across CPV segments gives statistic 23.45 ($p = 3.26e - 05$).

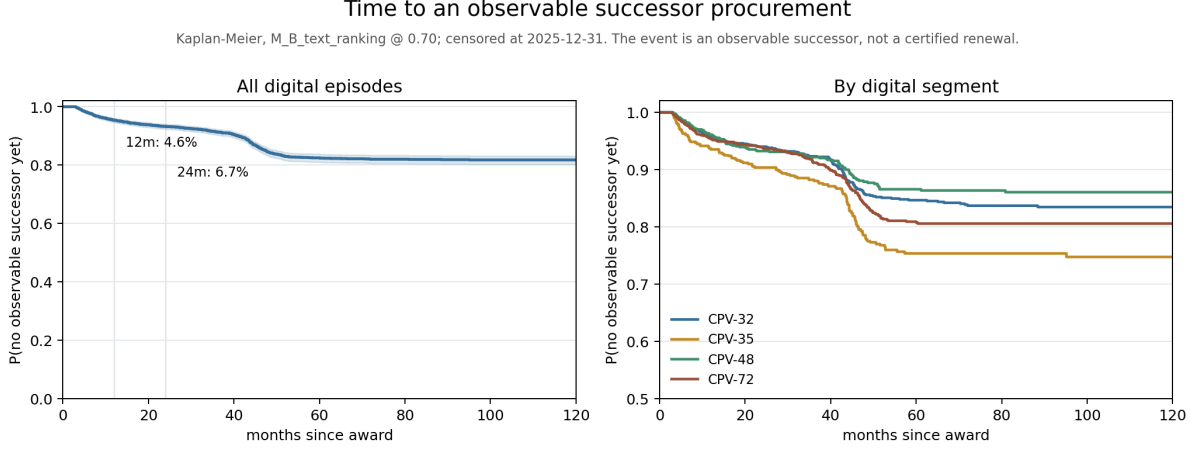


Figure 5: Kaplan–Meier survivor functions. The left panel carries the absolute level, which is linkage-conditioned and quoted only with that caveat; the right panel carries the segment ordering, which survives every sensitivity arm.

Cox model. The parsimonious model (8 covariates, 544 events) gives:

Covariate	HR	95% CI	p
framework_flag	1.751	[1.435, 2.136]	3.38e-08
has_validated_siren	1.082	[0.885, 1.323]	0.443
award_year_centered	1.107	[1.066, 1.149]	1.21e-07
digital_segment_CPV-35	1.553	[1.218, 1.981]	0.000383
digital_segment_CPV-48	0.828	[0.638, 1.073]	0.153
digital_segment_CPV-72	1.056	[0.850, 1.310]	0.624
buyer_region_Normandie	0.800	[0.640, 1.000]	0.0501
buyer_region_Pays de la Loire	1.003	[0.815, 1.234]	0.979

Table 3: Cox hazard ratios, main linkage arm. In-sample C-index 0.626.

Framework-agreement episodes and CPV-35 carry the largest hazard ratios among substantively interpretable covariates. These are descriptive associations with the observed hazard, not causal effects.

Covariate	Test statistic	p	PH assumption
award_year_centered	70.733	4.09e-17	violated
buyer_region_Normandie	0.608	0.436	ok
buyer_region_Pays de la Loire	2.455	0.117	ok
digital_segment_CPV-35	0.041	0.84	ok
digital_segment_CPV-48	0.671	0.413	ok
digital_segment_CPV-72	3.723	0.0537	ok
framework_flag	7.040	0.00797	violated
has_validated_siren	6.763	0.00931	violated

Table 4: Schoenfeld-residual proportional-hazards tests.

Proportional-hazards diagnostic. The assumption is rejected ($p < 0.05$) for `award_year_centered`, `framework_flag`, `has_validated_siren`. These coefficients are therefore reported as time-averaged associations rather than constant hazard ratios; stratification or time-interaction terms would be the next refinement if individualized prediction were required.

Temporal validation. Training on 2015-2021 (2,470 episodes, 392 events) and evaluating on the guideline-aligned 2022-2024 window (1,004 episodes, 107 events) gives C-index 0.606 in-sample versus 0.479 out-of-time. Extending the test window to 2022-2025 (1,330 episodes, 152 events), without refitting, gives 0.518. Both are close to the 0.5 chance line: individualized out-of-time discrimination is weak. The two windows also differ in follow-up, and [Uno et al. \(2011\)](#) show that the concordance statistic for right-censored data converges to a quantity depending on the censoring distribution, so the gap between the two figures should not be read as a change in discriminative ability. That is a result, not a prompt to retune – the model is retained as a descriptive risk-factor summary and nothing in the operational deliverable rests on it. Part of the gap is structural, because episodes awarded from 2022 onwards can contribute only short-gap events.

Borderline-link robustness. Dropping the 280 episodes whose best candidate scores within ± 0.05 of the acceptance threshold removes 133 events and leaves the direction of both headline hazard ratios unchanged (CPV-35 1.55 to 1.78; framework 1.75 to 1.62). The absolute Kaplan–Meier level falls, from 4.6% to 3.7% at 12 months, which is the mechanical consequence of removing borderline events. The comparative conclusions therefore do not rest on borderline linkage decisions, while absolute probabilities remain threshold-uncertain.

Model	Parameters	Log-likelihood	AIC	BIC
GeneralizedGamma	3	-3757.5	7521.0	7539.7
LogNormal	2	-3784.6	7573.2	7585.7
LogLogistic	2	-3807.3	7618.6	7631.1
Weibull	2	-3814.7	7633.4	7645.8
Exponential	1	-3829.7	7661.3	7667.6

Table 5: Parametric survival model comparison.

Parametric models. `GeneralizedGamma` has the lowest AIC and BIC and was checked graphically against the Kaplan–Meier curve. It is reported as the best-fitting family and as the instrument any extrapolation past the 2025-12-31 cutoff would use. It is *not* the source of the exported 12/24-month conditional successor probabilities (`survival_conditional_probabilities.csv`): those come from the Kaplan–Meier estimator, each with a 500-draw episode-bootstrap interval. The reason is visible in the fit itself – around age 36 months, just before the observed renewal shoulder, every parametric family including the selected one reports roughly 0.02–0.03 for the next twelve months where Kaplan–Meier reports near 0.075. Since every horizon this report quotes falls inside the observed window, the extrapolation that would justify accepting that smoothing is not needed. These are estimated probabilities of an *identifiable observable successor procurement*, not certified renewal probabilities, and no active Gigalis portfolio was available to score, so they cover the study cohort rather than a live deployment set.

Operational 12- and 24-month probabilities. This is the quantity the business question actually asks for. For an episode that has reached age a months with no accepted successor, the probability that one becomes visible within the next h months is $P(T \leq a + h \mid T > a) = 1 - S(a + h)/S(a)$, read off the Kaplan–Meier estimator with 500-draw episode-bootstrap intervals.

Age at assessment	$P(12m)$	95% CI	$P(24m)$	95% CI
0 months	4.62%	[3.91, 5.24]	6.73%	[5.91, 7.58]
12 months	2.21%	[1.70, 2.73]	4.26%	[3.55, 4.98]
24 months	2.09%	[1.58, 2.60]	9.40%	[8.33, 10.68]
36 months	7.46%	[6.52, 8.59]	9.69%	[8.58, 11.12]
48 months	2.41%	[1.76, 3.07]	2.89%	[2.15, 3.72]

Table 6: Conditional probability that an observable successor appears within the next 12 or 24 months, by current episode age.

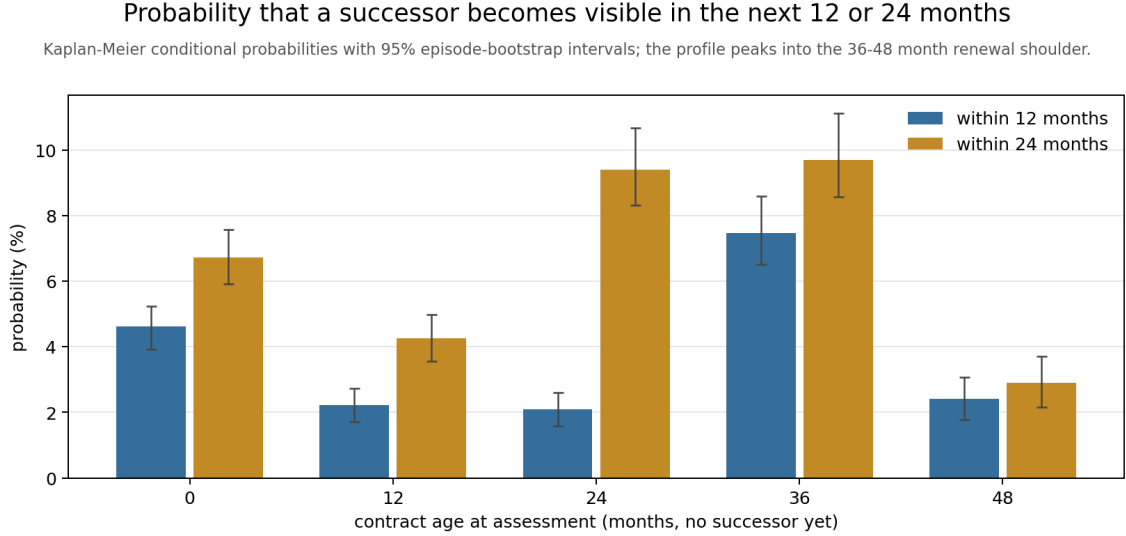


Figure 6: The same quantities with their bootstrap intervals. The profile is not monotone in age: it rises into the 36–48 month renewal shoulder and falls away after it, which is the empirical feature every parametric family smooths out.

The intervals are wide relative to the estimates, and these numbers rank ages and segments rather than calibrating individual forecasts. They estimate an observable successor procurement appearing in BOAMP, not a certified renewal, and they are conditional on the linkage rule.

Template-risk robustness. The four linkage arms and the borderline band both perturb where the acceptance bar sits. Neither addresses the false-positive mechanism identified above, because that mechanism produces links well *above* the bar: shared framework boilerplate can drive the character analyser high between unrelated objects, and because M_B ranks candidates within each anchor independently, one such episode can be accepted for several anchors. A stricter threshold does not remove either signature and can enrich for them. Two observable signatures, both published by the candidate-generation audit, define the at-risk group: word-level similarity below 0.50 (65 links) or a successor episode shared with another anchor (127 links), flagging 173 of the 544 accepted links (31.8%). Those anchors are re-censored at the cutoff rather than dropped, because a spurious link means the anchor had no observed successor and should still contribute its full follow-up as censored exposure.

Analysis	Episodes	Events	KM 12m	KM 24m	CPV-35 HR	Framework HR
Main	3,800	544	4.62%	6.73%	1.553	1.751
Re-censoring risk	3,800	371	2.64%	3.88%	1.541	1.692

Table 7: Headline results with template-risk links re-censored.

This is the check the framework-agreement finding most needs, since framework boilerplate is the text driving the mechanism: were the higher framework hazard an artefact of shared legal wording, re-censoring would collapse it. Both headline hazard ratios keep their side of 1 and move little, so the comparative findings are not products of the documented false-positive mechanism. The absolute Kaplan–Meier level falls by roughly the share of events re-censored, which is arithmetic rather than evidence. The check bounds the mechanism’s influence; it does not establish that the flagged links are false, and most of them are not.

Detectability and selection diagnostic. Comparing linked and censored episodes on standardized mean differences, the largest gap is `text_length_chars` (SMD 0.262). This indicates possible differential detectability across episode characteristics, not proof of causal linkage bias; it cannot be fully separated from genuine heterogeneity in renewal behaviour using BOAMP alone.

Linkage sensitivity. Event counts range from 296 to 1,332 across the four retained linkage arms, so absolute probabilities are linkage-sensitive by construction. Cox effects under the same four arms:

Covariate	Strict	Main	Looser	High-recall	Robustness
award_year_centered	1.017	1.107	1.136	1.101	Robust
buyer_region_Normandie	0.803	0.800	0.794	0.726	Mostly Robust
buyer_region_Pays de la Loire	0.977	1.003	1.023	0.891	Linkage Sensitive
digital_segment_CPV-35	1.792	1.553	1.488	1.484	Robust
digital_segment_CPV-48	0.652	0.828	0.807	0.885	Mostly Robust
digital_segment_CPV-72	1.108	1.056	1.036	1.557	Linkage Sensitive
framework_flag	1.281	1.751	1.953	1.623	Robust
has_validated_siren	1.193	1.082	1.107	1.650	Linkage Sensitive

Table 8: Cox hazard ratios across linkage-definition sensitivity arms.

Framework flag, CPV-35, and centered award year are the most robust associations; buyer-region and CPV-72 effects are linkage-sensitive and should not be over-interpreted.

15 Trend Results

Segment	Direction	Slope/qtr	p	Last stable PELT break	HMM regime
Overall	stable_or_uncertain	-0.11	0.921	–	growth
CPV-32	stable_or_uncertain	-0.01	0.989	2020Q2	growth
CPV-35	stable_or_uncertain	0.03	0.923	–	–
CPV-48	decreasing	-0.84	0.032	2024Q1	–
CPV-72	stable_or_uncertain	0.70	0.285	2021Q1	growth

Table 9: Trend signal matrix: OLS 12-quarter slope, PELT breaks, and HMM current regime (Overall and top-2 segments only).

CPV-48 is the only segment with a statistically distinguishable 12-quarter decline at the exploratory $\alpha = 0.10$ level; the rest are **stable_or_uncertain** by this signal.

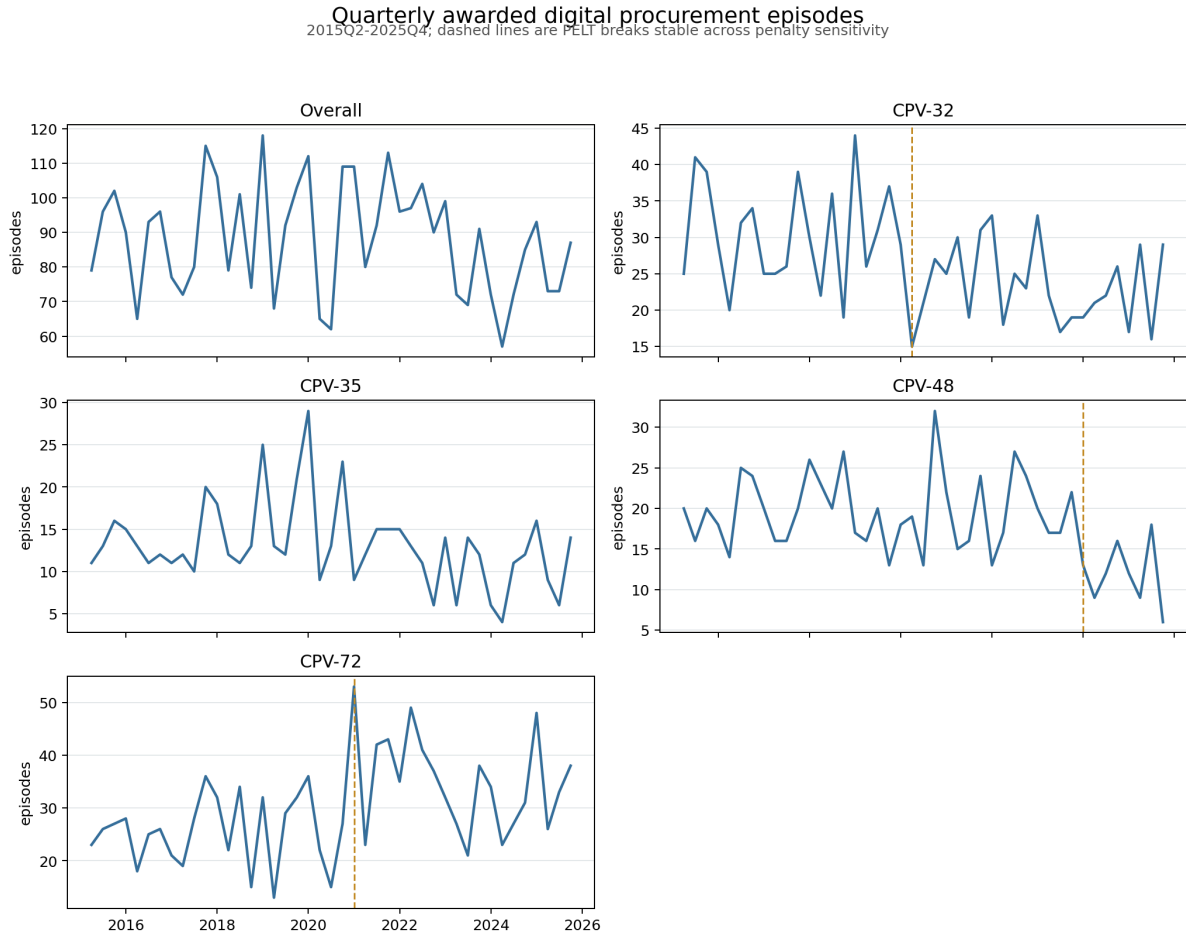


Figure 7: Quarterly awarded digital procurement episodes, 2015Q2–2025Q4. Dashed lines mark PELT breaks that survive all three penalty multipliers. A break dates a level shift; it does not explain one.

Operational reading. Each segment’s signals translate into a monitoring action, carried in the **business_recommendation** column of **trend_signal_matrix.csv** so that this text and the materialised table cannot drift apart. These are readings of descriptive evidence, not forecasts and not causal explanations: a PELT break dates a level shift without explaining it, and none of the statements below should be quoted as attributing a shift to policy, COVID, regulation, or technology.

- **Overall** (stable_or_uncertain): Maintain monitoring for Overall; no statistically clear recent direction at the pre-declared exploratory level. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.
- **CPV-32** (stable_or_uncertain): Maintain monitoring for CPV-32; no statistically clear recent direction at the pre-declared exploratory level. A penalty-stable level shift is dated 2020Q2; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.

- **CPV-35** (stable_or_uncertain): Maintain monitoring for CPV-35; no statistically clear recent direction at the pre-declared exploratory level.
- **CPV-48** (decreasing): Investigate the recent decline in CPV-48 before reducing or expanding procurement capacity; confirm whether it reflects demand, publication practice, or a routing change to another channel. A penalty-stable level shift is dated 2024Q1; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause.
- **CPV-72** (stable_or_uncertain): Maintain monitoring for CPV-72; no statistically clear recent direction at the pre-declared exploratory level. A penalty-stable level shift is dated 2021Q1; treat it as a break candidate to be explained with documentary evidence, not as a demonstrated cause. The HMM currently reads this series as ‘growth’, which describes recent quarter-over-quarter change and need not agree with the 12-quarter slope.

Segment	ADF (H0: unit root)	KPSS (H0: level stationary)	Note
Overall	stat=-5.890, p=0.000	stat=0.268, p=0.100	
CPV-32	stat=-6.086, p=0.000	stat=0.808, p=0.010	
CPV-35	stat=-1.684, p=0.439	stat=0.444, p=0.058	
CPV-48	stat=-1.749, p=0.406	stat=0.545, p=0.032	
CPV-72	stat=-5.439, p=0.000	stat=0.611, p=0.022	

Table 10: Stationarity diagnostics per segment.

ADF and KPSS test opposite null hypotheses and are read jointly; disagreement between them means the series is not cleanly classified as stationary or non-stationary over this 43-quarter window.

The HMM regime column above reports each covered segment’s current-quarter regime and posterior probability, fit on the quarter-over-quarter change in episode count rather than its level. Because the model is fit on noisy, low-count quarterly data, the **plateau** state is a data-driven middle tier rather than a change centered exactly at zero, and it is read alongside **decline/growth** rather than as "no change." The HMM’s current-quarter regime and the 12-quarter OLS slope are complementary diagnostics computed over different windows; they are not required to agree, and this report does not adjust either to match the other.

16 Final Linkage Specification

The final event definition is **M_B_text_ranking @ 0.70**, a frozen conservative observable-successor baseline, with the stricter ($M_B@0.80$), looser ($M_B@0.60$), and weighted-gated ($M_C@0.70$) variants as the required sensitivity analyses, together with the borderline-band check. The threshold is not claimed to be optimal; it is claimed to be pre-specified. It was fixed before the regional reference was consulted and has not been moved since, so the locked-split figures are held out rather than fitted. Moreover, the completed model-assisted review confirmed only 14 of 20 sampled production links at 0.70 conservatively, so lowering an unreviewed threshold would not support a stronger accuracy claim. This is a precision-first design. Low recall is accepted as an explicit trade-off, but the observed event rate is not a mathematical lower bound: missed successors push it downward while residual false links can push it upward. [Doidge and Harron \(2019\)](#) make the general point that missed and false links misclassify in opposite directions, which is why no one-sided bound follows from a linkage-conditioned event rate. It is therefore a linkage-conditioned indicator whose absolute value must be read with the strict, looser, and weighted-gated sensitivity results and the borderline-band check. Independent specialist review

is required only before claiming externally validated precision or promoting a new threshold, not to complete this linkage-conditioned descriptive study. If that review shows weaker precision, the claim should be narrowed further rather than forcing a more complex model.

Segment continuity of the accepted links. Because hard same-CPV blocking is not imposed, the accepted links are checked after the fact for segment drift. Of the 544 accepted primary links, 538 have a CPV division observed on both sides; 351 (65.2%) stay inside one division and 187 (34.8%) cross divisions, while 75.8% share at least one division across their full CPV lists. The comparison that interprets this is the reviewed reference, which crosses divisions at 39.1% – close to the 34.8% observed among accepted links. Relaxed CPV blocking therefore reproduces roughly the segment-continuity behaviour the reviewed successors already display, rather than introducing drift of its own. The largest cross-division flows are the goods-to-services pairs named above. Full transition counts are in `data/processed/boamp/candidate_generation_cpv_transitions.csv`.

Inspecting the cross-division links by hand also identifies the dominant false-positive mechanism, which is textual rather than categorical. French award notices carry long standardised framework-agreement boilerplate – *accord-cadre mixte*, *bons de commande*, *bordereau des prix unitaires*, *marchés subséquents* – and for buyers whose notices are dominated by that template, character-level similarity can clear the threshold on shared legal phrasing alone. 65 accepted links carry word-level similarity below 0.50 and so rest on the character analyser; inspection shows most are legitimate rewordings of the same object, but this is also the band in which the failure mode lives. Because M_B ranks candidates within each anchor independently, with no one-to-one constraint, one episode can be accepted for several anchors: the most reused successor here is accepted by 11 anchors, and 44 of the 461 distinct successors serve more than one. Multi-lot programmes make much of that legitimate, but the clearest spurious instance found by inspection is a single catering framework accepted as the successor of nine unrelated digital anchors from one regional authority, whose notices are almost entirely template text. A hard same-CPV block would suppress that particular case but would not fix the mechanism, which also operates within a division, and would cost the 9 genuine reviewed successors quantified above. The mechanism is therefore documented as a known contributor to the measured imprecision rather than patched, and it is one reason the absolute event level is reported as linkage-sensitive.

17 Limitations And Robustness

- BOAMP does not consistently encode legal renewal status; accepted links are observable successor procurements, not legal renewal proof.
- The reference labels are a spot-checked LLM research pass, not anchor-by-anchor verification and not an independent specialist panel, and its negatives are corpus-relative, so the reported false-positive rate is a sample diagnostic rather than a population-wide rate (§4).
- The held-out reference is small: 18 positive anchors and 8 accepted M_B links. The selected threshold should not be over-tuned and its accuracy remains provisional.
- Buyer standardisation is improved but remains an important risk area. The legal-form audit is retained to catch name-only and cross-legal-form cases.
- Missing duration and expiry fields are not imputed. This avoids creating false timing certainty.
- The 3-state HMM regime labels are fit on noisy, short (43-quarter) count series; state-count and window choices are not exhaustively validated, and a ‘plateau’ label is a relative middle tier rather than a change centered at zero.

- Technology segmentation uses CPV divisions, not a learned taxonomy (§8); this is a real scope reduction, disclosed rather than concealed.

18 Perspectives

18.1 Causal Inference (Outline Only)

The internship guide frames a distinct causal question, separate from this report’s descriptive linkage/survival/trend results: does a Gigalis pooled framework agreement actually change member purchasing behaviour (amounts, frequency, segments), or does Gigalis simply capture demand that would have existed anyway? **This question is not answered here, and no causal estimate is computed in this repository.** Answering it requires data this project’s BOAMP-only corpus does not contain: which buyers are Gigalis members, and when each joined. Attempting a substitute causal claim from BOAMP alone – for example treating CPV-segment membership or region as a proxy treatment – would confound procurement domain and geography with Gigalis adoption, and this project has consistently avoided exactly this kind of unsupported inferential leap elsewhere (survival hazard ratios are reported as associations, not effects, for the same reason).

If Gigalis-internal membership and adoption-date data were joined to this corpus, the natural identification strategy is a **staggered-adoption difference-in-differences** design ([Callaway and Sant’Anna, 2021](#)), because members join the central purchasing body at different calendar dates rather than simultaneously – the specific setting that motivates that estimator over classical two-period DiD. The parallel-trends assumption (comparable purchasing trajectories absent Gigalis membership) would need to be assessed with pre-adoption trend plots per member cohort. Propensity-score matching ([Angrist and Pischke, 2009](#)) is a secondary candidate if adoption is not sharply timed but depends on observable buyer characteristics (size, budget, sector); regression discontinuity would apply only if a hard eligibility threshold (e.g. a population cutoff) determines Gigalis access, which is not currently known to be the case. Any of these designs should state its identifying assumptions as an explicit causal graph ([Pearl, Glymour and Jewell, 2016](#)) before estimation, not after.

This section is a design outline for future work conditional on Gigalis supplying membership data, not a promise that such data will become available within this internship.

19 Interpretation And Scope Of Claims

The technical report should defend the project as a reproducible measurement pipeline, not as a black-box prediction system. The strongest statement is:

Because legal renewal status is not directly observed in BOAMP, the study estimates observable successor procurements. Candidate generation restricts comparison to the same buyer and a plausible future interval; the selected method then accepts only the strongest textually similar candidate above a fixed threshold. This prioritises precision over recall, which is appropriate because false positive links would create artificial survival events.

20 Reproducibility Artifacts

- `data/processed/boamp/linkage_evaluation_dev.json`
- `data/processed/boamp/linkage_evaluation_validation.json`
- `data/processed/boamp/regional_benchmark/modeling/modeling_summary.json`

- `data/processed/boamp/regional_benchmark/regional_benchmark_manifest.json`
- `data/reference/regional_link_benchmark/`
- `data/processed/boamp/survival_dataset_summary.json`
- `data/processed/boamp/linkage_candidates_summary.json`
- `data/processed/boamp/survival_analysis_summary.json`
- `data/processed/boamp/survival_cox_results.csv`
- `data/processed/boamp/survival_ph_diagnostics.csv`
- `data/processed/boamp/survival_parametric_comparison.csv`
- `data/processed/boamp/survival_cox_linkage_sensitivity.csv`
- `data/processed/boamp/trend_analysis_summary.json`
- `data/processed/boamp/trend_signal_matrix.csv`
- `DATA_QUALITY_REPORT.md`
- `TREND_ANALYSIS_REPORT.md`
- `SURVIVAL_ANALYSIS_REPORT.md`
- `PROJECT_WORK_PROTOCOL.md`
- `REVIEW_AUDIT_RESULTS.md`
- `EXECUTIVE_SUMMARY.md`

21 Methodological References

The source and identity definitions follow the official BOAMP and INSEE documentation. CPV interpretation follows Commission Regulation (EC) No 213/2008. TF-IDF cosine similarity follows the standard vector-space definition. Record linkage, survival estimation, proportional-hazards diagnostics, and change-point detection are supported by the original Fellegi–Sunter (1969), Davis–Goadrich (2006), Saito–Rehmsmeier (2015), Kaplan–Meier (1958), Cox (1972), Grambsch–Therneau (1994), and PELT (Killick et al., 2012) methods. Regime detection follows the Markov-switching framework of Hamilton (1989). The causal-inference outline in §18.1 draws on Angrist and Pischke (2009), Pearl, Glymour and Jewell (2016), Athey and Imbens (2017), and Callaway and Sant’Anna (2021). None of these sources validates this project’s own labels, numerical results, or thresholds; they justify method choices only. Full URLs and the specific design implications are recorded in `METHODOLOGICAL_REFERENCES.md`.

- [Official BOAMP API \(DILA\)](#).
- [INSEE SIREN definition](#) and [INSEE SIRET definition](#).
- [Commission Regulation \(EC\) No 213/2008 on CPV](#) and [Directive 2014/24/EU, Article 33](#).
- [Cosine similarity documentation](#).
- [Davis–Goadrich \(2006\)](#) and [Saito–Rehmsmeier \(2015\)](#) on ROC and precision-recall evaluation for skewed binary decisions.

- [Fellegi–Sunter \(1969\)](#), [Kaplan–Meier \(1958\)](#), and [Cox \(1972\)](#).
- [Grambsch–Therneau \(1994\)](#) and [Killick et al. \(2012\)](#).
- [Hamilton \(1989\)](#) on Markov-switching regime models, the framework underlying the trend HMM.
- [Angrist and Pischke \(2009\)](#), [Pearl, Glymour and Jewell \(2016\)](#), [Athey and Imbens \(2017\)](#), and [Callaway and Sant’Anna \(2021\)](#) on causal identification, cited only to support the outlined-not-executed causal-inference perspective.